

NLP — 10-Page Master Quick Revision

High-Yield Formula Sheet, Architecture Matrices & BIT Mesra PYQ Solutions

🧠 10-Page Master Quick Revision — Natural Language Processing (CS24351)

Page 1-2: Linguistic Levels, BPE Tokenization & Regex Extraction

Page 5-6: TF-IDF, Word2Vec (CBOW/Skip-gram), GloVe & FastText

Page 9-10: BERT vs. GPT, BLEU Score Formulation & AI Ethics

Page 3-4: N-Gram Language Models, Perplexity & Smoothing Formulas

Page 7-8: LSTM Gating, Scaled Dot-Product & Multi-Head Attention

⚡ High-Yield NLP Formula Master Matrix

Model / Metric	Exact Formula / Rule	Key Insight
TF-IDF	$\text{TF-IDF} = \text{TF}(t, d) \times \log \left(\frac{ D }{ \{d:t \in d\} } \right)$	Balances local frequency with global specificity.
Word2Vec SGNS	$\vec{v}(\text{King}) - \vec{v}(\text{Man}) + \vec{v}(\text{Woman}) \approx \vec{v}(\text{Queen})$	Linear semantic vector offsets.
Scaled Attention	$\text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$	$\sqrt{d_k}$ prevents vanishing softmax gradients.
Perplexity	$\text{PP}(W) = 2^{-\frac{1}{N} \sum \log_2 P(w_i w_{i-1})}$	Lower perplexity indicates better predictive model.